

numbers of lines that are created and destroyed then gives

$$2I + E = \sum_i n_i V_i, \quad (6.3.11)$$

where  $I$  and  $E$  are the numbers of internal and external lines,  $V_i$  are the numbers of vertices of various types labelled  $i$ , and  $n_i$  are the number of lines attached to each vertex. (This also holds separately for fields of each type.) In particular, if all interactions involve the same number  $n_i = n$  of fields, then this reads

$$2I + E = nV, \quad (6.3.12)$$

where  $V$  is the total number of all vertices. In this case, we can eliminate  $I$  from Eqs. (6.3.4) and (6.3.11), and find that for a connected (i.e.,  $C = 1$ ) graph the number of vertices is given by

$$V = \frac{2L + E - 2}{n - 2}. \quad (6.3.13)$$

For instance for a trilinear interaction the diagrams for a scattering process ( $E = 4$ ) with  $L = 0, 1, 2, \dots$  has  $V = 2, 4, 6, \dots$  vertices. In general, the expansion in powers of the coupling constants is an expansion in increasing numbers of loops.

## 6.4 Off the Mass Shell

In the Feynman diagrams for any  $S$ -matrix element all external lines are 'on the mass shell'; that is, the four-momentum associated with an external line for a particle of mass  $m$  is constrained to satisfy  $p_\mu p^\mu = -m^2$ . It is often important also to consider Feynman diagrams 'off the mass shell', for which the external line energies like the energies associated with internal lines are free variables, unrelated to any three-momenta. For one thing, these arise as parts of larger Feynman diagrams; for instance, a loop appearing as an insertion in some internal line of a diagram could be regarded as a Feynman diagram with two external lines, both off the mass shell.

Of course, once we calculate the contribution of a given Feynman diagram off the mass shell, it is easy to calculate the associated  $S$ -matrix elements by going to the mass shell, taking the four-momentum  $p^\mu$  flowing along the line *into* the diagram to have  $p^0 = \sqrt{\mathbf{p}^2 + m^2}$  for particles in the initial state and  $p^0 = -\sqrt{\mathbf{p}^2 + m^2}$  for particles in the final state, and including the appropriate external line factors  $(2\pi)^{-3/2}u_\ell$  or  $(2\pi)^{-3/2}v_\ell^*$  for initial particles or antiparticles and  $(2\pi)^{-3/2}u_\ell^*$  or  $(2\pi)^{-3/2}v_\ell$  for final particles or antiparticles. Indeed, when we come to the path integral approach in Chapter 8 we shall find it easiest first to derive the Feynman